

02 - Singly Linked Lists

Due Sep 22 at 2pm **Points** 1 **Questions** 5 **Available** until Sep 22 at 2pm
Time Limit 30 Minutes **Allowed Attempts** 10

This quiz was locked Sep 22 at 2pm.

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	7 minutes	1 out of 1

❗ Correct answers are hidden.

Score for this attempt: 1 out of 1

Submitted Sep 15 at 2:51pm

This attempt took 7 minutes.

Question 1

0.2 / 0.2 pts

A one-way list with a header node (H1L) is given. The pointer H points to the header of the list. The list contains integers, and it is unordered. It is also given an integer k. Delete from the list all occurrences of k except the first.

Complete the missing steps of the following pseudocode:

del_k(H:E1*; k:Z)

found := false; [Select]

while [Select]

if p->key = k

if found

pp->next:=p->next

delete p

else

found := true

[Select]

**end if****else****pp := p****end if**

[Select]

**end while**

Answer 1:

pp:=H; p:=H->next

Answer 2:

p ≠ 0

Answer 3:

pp->next:=p->next

Answer 4:

pp:=p

Answer 5:

p:=pp->next

Question 2**0.1 / 0.1 pts**

The pointer "p" points to an arbitrary element in a simple one-way linked list (S1L). The pointer "r" points to an element of type E1. We want to insert the element (*r) after the element (*p) in the list. Choose the correct solutions.

✓ (*r).next:=p->next; (*p).next:=r;

- ☐ (*r).next:=p; (*p).next:=r ;
- ☐ p→next:=r; r→next:=p→next;
- ☒ r→next:=p→next; p→next:=r;
- ☐ (*r)→next:=(*p).next; p→next:=r;

Question 3

0.3 / 0.3 pts

Given a one-way list with a header node (H1L), the pointer H points to the header. The list contains integers and is unordered. Determine what the algorithm below does and then what the result would be for the lists given below.

```

what_is_this(H:E1*)
  pp := H; p := H->next
  while p ≠ 0 and p->key>=0
    pp := p
    p := p->next
  end while
  pn := pp
  while p ≠ 0
    if p->key < 0
      pp := p
      p := p->next
    else
      pp->next := p->next
      p->next := pn->next
      pn->next := p
      pn := p
      p := pp->next
    end if
  end while

```

Specify what the result would be by running the algorithm on the lists below.

Give the elements separated by commas. Do not use spaces. For example:
3,0,-1

H ~ < 9,4,-1,-3 > The elements of the resulting list:

9,4,-1,-3

$H \sim \langle -3, -1, 2 \rangle$ The elements of the resulting list:

$H \sim \langle 1, -2, 5, 6, -3, 4, 0, -5 \rangle$ The elements of the resulting list:

Answer 1:

Answer 2:

Answer 3:

Question 4

0.3 / 0.3 pts

L points to the beginning of a simple one-way list (S1L). We want to move the first element to the end of the list. No changes are needed if the list is empty or contains only one element.

Complete the missing steps of the following pseudocode:

First2Last()

if not

p:=L

q:=L

while

q:=q->next

end while

[Select]



end if

Answer 1:

$\&L : E1^*$

Answer 2:

$L = 0 \vee L \rightarrow \text{next} = 0$

Answer 3:

$L := p \rightarrow \text{next}$

Answer 4:

$q \rightarrow \text{next} \neq 0$

Answer 5:

$q \rightarrow \text{next} := p$

Answer 6:

$p \rightarrow \text{next} := 0$

Question 5

0.1 / 0.1 pts

Which of the following statements are true for a simple one-way list (S1L)?



Accessing any i -th element of the list can be done in constant time. ($1 \leq i \leq n$, where n is the number of elements in the list.)



The effect of $p \rightarrow \text{next} := p \rightarrow \text{next} \rightarrow \text{next}$ is that it unlinks the element after ($*p$) from the list. (We know that $p \neq 0$ and $p \rightarrow \text{next} \neq 0$.)



If we want to insert a new element before an element with address p in the list, the operation has a time complexity of $O(n)$, where n is the number of elements in the list. (Apart from the address of p , we only know the address of the first element of the list.)



If we want to delete any element from the list, the address of the previous element is always required.



The time complexity of accessing the last element of the list is $\Theta(n)$, where n is the number of elements in the list. (Only the address of the first element is known.)

Quiz Score: **1** out of 1